Universidad Nacional Abierta y a Distancia

Vicerrectoría Académica y de Investigación

Learning Guide for the Development of the Practical Component of the Course Forecasting

Code: 203238431

1. General Information about the Practical Component

Table 1. *General Information.*

Aspect	Description
1. Methodological strategy	Problem Based Learning
2. Course typology	Methodological
3. Evaluation moment	**Intermediate**
4. Activity score	**100**
5. Number of activities of the practical component registered in this guide	**1**
6. Independent work hours of the student	**56**
7. Hours of teaching support	**24**
8. Type of formative practice	**Simulated**

2. With this/these practical component activity/activities, it is expected that students achieve and demonstrate the following learning outcome(s):

The activity consists of constructing recurrent neural network (RNN) and long short-term memory (LSTM) models for time-series forecasting using Python and Keras. Students will apply theoretical concepts by implementing these models in a Jupyter notebook titled

3. **General description of the activity/activities of the practical component.**

Table 2. *Activity 1 Information.*

Aspect	Description
1. Scenarios of the practical component	**Simulated with ICT Support**
2. Type of activity	Independent
3. Activity number	3
4. Activity score	100 points
5. The activity starts on:	Wednesday, April 15, 2026
6. The activity ends on:	Tuesday, May 12, 2026

The following materials and resources are required for the development of this activity:

Problem Situation

Develop each of the exercises by taking into account each of the attached files and completing all the proposed points. The exercises involve different types of real-world data through which each of the concepts in this phase can be developed.

The activity consists of:
To develop this activity, follow these steps:

Exercise 1:

You have been provided with the dataset and code:

Appendix 1- Google_Colab.pdf

Appendix 2 - Beijing.csv

Appendix 3 - Introduction_to_Deep_Learning_student.ipynb

More specifically, you will do the following:

- Interpolate the missing values in the "PM_Nongzhanguan" series and plot the last 42 days of the series to get a feel for the data.

- Extract the train and test data for the "PM_Nongzhanguan" series using the function get_train_test_data. Use the following set of parameters:

  - series_days : 56
  - input_hours : 12
  - test_hours : 48

- For reference, below is how we called the function earlier on. You can also pull up the function documentation to review the various arguments.
- Using the extracted train data to fit a simple RNN and use the test data to generate and plot predictions.
- Start with a simple baseline - few cell units and epochs. From here, try to make the model more expressive by increasing units and epochs until you're satisfied with the model's predictions.
- Be careful not to set units and/or epochs too high. The model may become very slow to train and also start to badly overfit the training data with the extra complexity you've added.

Exercise 2:

Adapt the code above from Example 2 in order to build an LSTM that forecasts the "PM_Dongsi" series. Use the following set of parameters:

series_days : 50

input_hours : 12

test_hours : 48

Remember our simple parameter tuning strategy, and that LSTMs take significantly longer to train than simple RNNs.

Start with a simple baseline -- few cell units and epochs. From here, try to make the model more expressive by increasing units and epochs until you're satisfied with the model's predictions.

Be careful not to set units and/or epochs too high. The model may become very slow to train and also start to badly overfit the training data with the extra complexity you've added.

Exercise 3:

You have been provided one dataset and one code:

Appendix 4 - co2-ppm-mauna-loa-19651980.csv

Appendix 5 - LSTM_student.ipynb

- Use the MinMaxScaler to normalize the training and testing set betwee 0 and 1. Name the normalized outputs norm_train and norm_test.
- Write a function that separates the data into an input array (X) which has all the datapoint at times t-i's, and an output array which contains the data at timepoint t.
- **Improve the results above by trying the following: increase the number of epochs, increase the look_back value and modify the Architecture**

Evidence of independent work:
The individual work evidences to submit are:

- Participate in the forum corresponding to **Phase 2: Exploration** to engage academically with your tutor and groupmates on the topics discussed. This contribution is not the final work but a meaningful and significant input that includes well-reasoned arguments.
- Submit a consolidated report in **PDF format**. The report must include:
  - Cover page.
  - Development of Exercise 1, 2, and 3, explaining each step and displaying headers and graphics with Python results that are clear and appropriate for understanding.
  - A link to the developed project code (e.g., GitHub, Google Drive, or another repository) must be included in the report.
  - Conclusions
  - Bibliography

Evidence of group work:
No group work evidences are required for this activity.

4. General guidelines for the elaboration of evidences

For evidence developed Choose an option. Please consider the following guidelines:

All group members must participate with their contributions in the development of the activity.

In each group, one member must be chosen to be responsible for submitting the requested product in the environment indicated by the teacher.

Before submitting the requested product, ensure that it meets all the requirements specified in this activity guide.

Only include as authors of the submitted product the group members who participated with contributions during the time allocated for the activity.

Please note that all individual or group written products must comply with "spelling and mechanics" standards and the submission conditions defined in this activity guide.

Regarding the use of references, consider that the product of this activity must comply with the Choose an option standards.

In any case, comply with the referencing standards and avoid academic plagiarism. To do so, you can support yourself by reviewing your written products using the Turnitin tool available in the virtual campus.

Consider that in Agreement 029 of December 13, 2013, Article 99, the following actions are considered as offenses against academic order, among others: item e) "Plagiarism, that is, presenting as one's own the entirety or part of a work, paper, document, or invention created by another person. It also includes the use of false citations or references, or proposing citations where there is no match between the citation and the reference," and item f) "Reproducing or copying, for profit, educational materials or results of research products that have intellectual property rights reserved for the University."

The academic sanctions that the student will face are as follows:

a) In cases of proven academic fraud in the respective academic work or evaluation, the grade imposed will be zero points without prejudice to the corresponding disciplinary sanction.

b) In cases related to proven plagiarism in academic work of any nature, the grade imposed will be zero points, without prejudice to the corresponding disciplinary sanction.